

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 6: Due by 2:45pm on Tuesday March 5, 2024.

Submit to WyoCourses a Jupyter notebook (both ipynb and HTML files) .

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [35 points] Consider the boundary value problem (BVP) $y'' = 18y^2$ with the boundary constraints: $y(1) = 1/3$ and $y(2) = 1/12$.
 - (A). [5 points] Prove that $y(t) = t^{-2}/3$ solves the BVP.
 - (B). [5 points] Let $N = 4$ (i.e., $h = 1/4$) and write down the associated finite difference equations for y_0, y_1, y_2, y_3 and y_4 .
 - (C). [10 points] Derive the associated Jacobian matrix J , and solve the nonlinear system of equations using the Newton's method.
 - (D). [5 points] Make two plots. The first should compare the estimates y_i to the true solution $y(t_i)$ at times t_i , and the second should plot the errors versus t_i .
 - (E). [10 points] Repeat Problems 1C-1D with $N = 10$, and describe how the error changes with the larger N value.
2. [35 points] Consider the BVP $9y''(t) + \pi^2 y(t) = 0$ with the boundary constraints: $y(0) = -1$ and $y(3/2) = 3$.
 - (A). [5 points] Prove that $y(t) = 3 \sin(\pi t/3) - \cos(\pi t/3)$ solves the BVP.
 - (B). [10 points] Consider solving the BVP using the collocation method, and let $w(t) = \sum_{j=0}^N c_j t^j$ denote your approximate solution. Write down the system of coupled equations to solve for the constants c_j .
 - (C). [10 points] Let $N = 2$ and solve for the constants c_j using a linear solver and make 2 plots. First, plot the approximation $w(t)$ and the true solution $y(t)$ versus $t \in [0, 3/2]$. Second, plot the error $e(t) = |y(t) - w(t)|$ versus t .
 - (D). [10 points] Repeat Problem 2C, but use different choices for N , i.e., $N = 6$ and $N = 12$. How does the error vary with N ?
3. [30 points] Consider the BVP $y'' = 3y - 2y'$ with the boundary constraints: $y(0) = e^3$ and $y(1) = 1$.
 - (A). [5 points] Prove that $y(t) = e^{3-3t}$ solves the BVP.
 - (B). [10 points] Write down the associated B-spline Galerkin finite element equations for general N . That is, find a system to solve for the coefficients $c_0, c_1, \dots, c_N$. *Hint: Use the following property for linear B-splines:*
$$\langle \phi_i, \phi'_j \rangle = \int_a^b \phi_i(t) \phi'_j(t) dt = \begin{cases} 1/2 & , \quad j = i + 1 \\ -1/2 & , \quad j = i - 1 \\ 0 & , \quad \text{otherwise.} \end{cases}$$
 - (C). [15 points] Let $N = 4$ and solve for the constants c_j . Make 2 plots. First, plot the approximation $w(t)$ and the true solution $y(t)$ versus $t \in [0, 1]$. Second, plot the error $e(t) = |y(t) - w(t)|$ versus t .